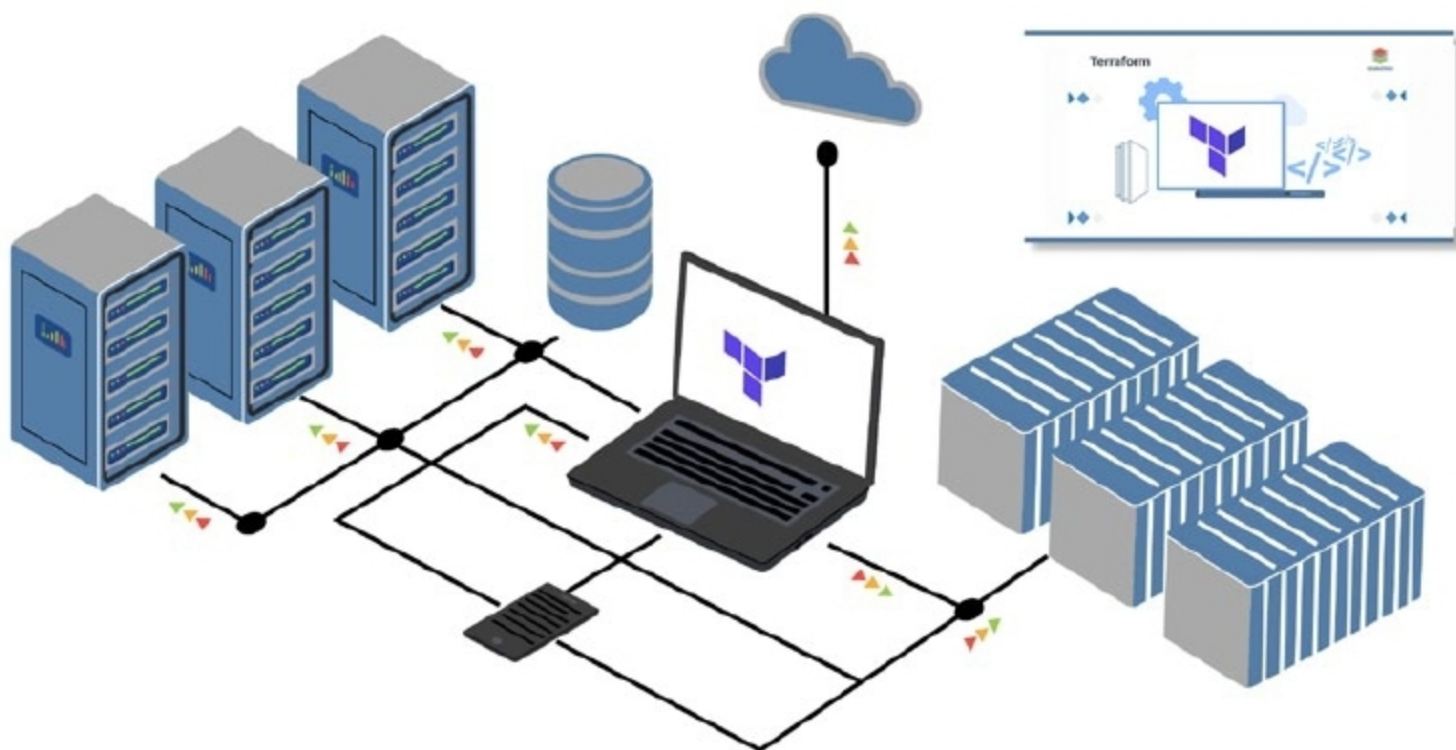


The Benefits of Using Terraform as a Tool for Infrastructure-as-Code (IaC)

DevOps tools have enabled software engineers to deploy application source code in a better way. Including Infrastructure-as-Code (IaC) in the DevOps cycle helps the transition to the cloud model, accommodating the shift from 'static' to 'dynamic' infrastructure.



The advent of DevOps helped minimise the dependence on sysadmins who used to set up infrastructure manually, while seated at some unknown corner of the office. Managing servers and services manually is not a very complicated task in a data centre. But when we move to the cloud, scale and start working with many resources from multiple providers (AWS, GCP, Azure, etc), manually setting up and configuring to achieve on-demand capacity slows things down. Being repetitive, the manual process is most likely error-prone. And it cannot be managed and automated when working with resources from different service providers together.

Infrastructure-as-Code (IaC)

This approach is the management and configuration of infrastructure (virtual machines, databases, load balancers and connection topology) in a descriptive cloud operating model. Infrastructure can be maintained just like application source code under the same version control. This lets the engineers maintain, review, test, modify and reuse their infrastructure and avoid direct dependence on the IT team. Systems can be deployed, managed and delivered fast, and automatically, through the IaC. There are many tools available for IaC such as CloudFormation (only for AWS), Terraform, Chef, Ansible and Puppet.

What is Terraform?

Terraform is an open source provisioning tool from HashiCorp (more can be read at <http://terraform.io/>) written in the Go language. It is used for building, changing and versioning infrastructure, safely and efficiently. Provisioning tools are responsible for the creation of server and associated services rather than configuration management (installation and management of software) on existing servers. Terraform acts as a provisioner, and focuses on the higher abstraction level of setting up the servers and associated services.

The infrastructure Terraform can manage includes low level components such as compute instances, storage,

